

# X5G: An Open, Programmable, Multi-Vendor, End-to-End, Private 5G O-RAN Testbed With NVIDIA ARC and OpenAirInterface

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**Abstract**—As Fifth generation (5G) cellular systems transition to softwarized, programmable, and intelligent networks, it becomes fundamental to enable public and private 5G deployments that are (i) primarily based on software components while (ii) maintaining or exceeding the performance of traditional monolithic systems and (iii) enabling programmability through bespoke configurations and optimized deployments. This requires hardware acceleration to scale the Physical (PHY) layer performance, programmable elements in the Radio Access Network (RAN) and intelligent controllers at the edge, careful planning of the Radio Frequency (RF) environment, as well as end-to-end integration and testing. In this paper, we describe how we developed the programmable X5G testbed, addressing these challenges through the deployment of the first 8-node network based on the integration of NVIDIA Aerial RAN CoLab Over-the-Air (ARC-OTA), OpenAirInterface (OAI), and a near-real-time RAN Intelligent Controller (RIC). The Aerial Software Development Kit (SDK) provides the PHY layer, accelerated on Graphics Processing Unit (GPU), with the higher layers from the OAI open-source project interfaced with the PHY through the Small Cell Forum (SCF) Functional Application Platform Interface (FAP). An E2 agent provides connectivity to the O-RAN Software Community (OSC) near-real-time RIC. We discuss software integration, network infrastructure, and a digital

twin framework for RF planning. We then profile the performance with up to 4 Commercial Off-the-Shelf (COTS) smartphones for each base station with iPerf and video streaming applications, as well as up to 25 emulated User Equipments (UEs), measuring a cell rate higher than 1.65 Gbps in downlink and 143 Mbps in uplink.

**Index Terms**—Private 5G, multi-vendor, GPU acceleration, O-RAN.

## I. INTRODUCTION

THE evolution of the Radio Access Network (RAN) in 5G networks has led to key performance improvements in cell and user data rates, now at hundreds of Mbps on average, and in air interface latency [2], thanks to specifications developed within the 3rd Generation Partnership Project (3GPP). From an architectural point of view, 5G deployments are also becoming more open, intelligent, programmable, and based on software [3], through activities led by the O-RAN ALLIANCE, which is developing the network architecture for Open RAN. These elements have the potential to transform how we deploy and manage wireless mobile networks [4], leveraging intelligent control, with RAN optimization and automation exercised via closed-loop data-driven control; softwarization, with the components of the end-to-end protocol stack defined through software rather than with dedicated hardware; and disaggregation, with the 5G RAN layers distributed across different network functions, i.e., the Central Unit (CU), the Distributed Unit (DU), and the Radio Unit (RU).

Open and programmable networks are often associated with lower capital and operational expenditures, facilitated by the increasing robustness and diversity of the telecom supply chain [5], now also including open-source projects [6], [7] and vendors focused on specific components of the disaggregated RAN. This, and increased spectrum availability in dedicated or shared bands, has opened opportunities to deploy private 5G systems, complementing public 5G networks with more agile and dynamic deployments for site-specific use cases (e.g., events, warehouse automation, industrial control, etc).

While the transition to disaggregated, software-based, and programmable networks comes with significant benefits, there are also several challenges that need to be addressed before

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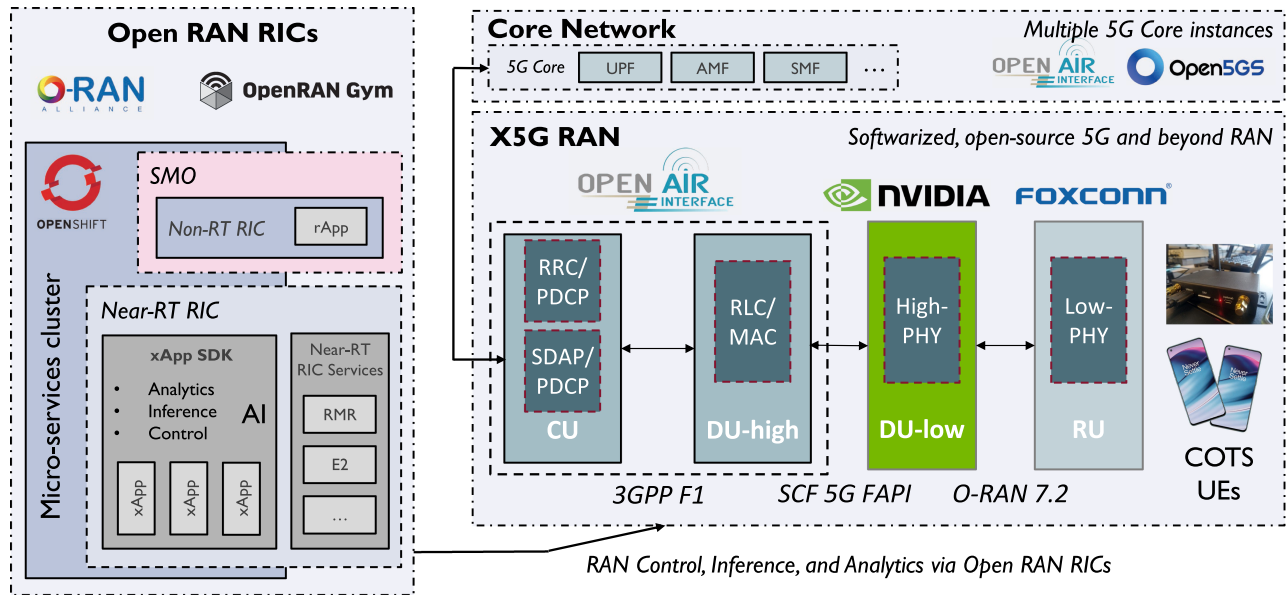


Fig. 1. X5G end-to-end programmable testbed overview.

Open RAN systems can align their performance or improve over traditional cellular systems. First of all, the radio domain still exhibits a low degree of automation and zero-touch provisioning for the RAN configuration, complicating the successful deployment of end-to-end cellular systems. Second, the diverse vendor ecosystem comes with challenges related to interoperability and end-to-end integration across several products, potentially from different vendors [8], [9], [10]. Third, the Digital Signal Processing (DSP) at the Physical (PHY) layer of the stack is a computationally complex element, using about 90% of the available compute when run on general-purpose CPUs, and thus introducing a burden on the software-based and virtualized 5G stack components. Finally, there are still open questions in terms of how the intelligent and data-driven control loops can be implemented with Artificial Intelligence (AI) and Machine Learning (ML) solutions that generalize well across a multitude of cellular network scenarios [11]. These challenges call for a concerted effort across different communities (including hardware, DSP, software, DevOps, AI/ML) that aims to design and deploy open, programmable, multi-vendor cellular networks and testbeds that can support private 5G requirements and use cases with the stability and performance of production-level systems.

In this paper, we introduce X5G, a private 5G network testbed deployed at Northeastern University in Boston, MA, and based on multiple programmable and open-source components from the physical layer all the way up to the Core Network (CN), as shown in Fig. 1. We discuss in detail the integration of a PHY layer implemented on Graphics Processing Unit (GPU) (i.e., NVIDIA Aerial) with OpenAirInterface (OAI) for the higher layers of the 5G stack [12]. This integration is based on the Small Cell Forum (SCF) Functional Application Platform Interface (FAPI), which regulates the interaction between the PHY and Medium Access Control (MAC) layers. This paper extends our recent work [1] by introducing new Open RAN elements and experimental results, including: (i) the integration of a near-real-time RAN Intelligent Controller (RIC) from the O-RAN Software Community (OSC) on an OpenShift cluster;

(ii) the validation of additional CNs, such as Open5GS and a commercial core from A5G; (iii) enhancements to the hardware architecture, including a more robust networking infrastructure and additional RAN servers; (iv) the evaluation of X5G under diverse operational conditions, such as stress testing its performance ensuring reliability for a Private 5G (P5G) network; and (v) a more comprehensive related work section.

X5G leverages the inline acceleration of demanding PHY tasks on GPU, hardware that is well equipped with massive parallelization of DSP operations, enabling scalability and the embedding of AI/ML in the RAN. The X5G infrastructure is continuously expanding through the integration of an increasing number of components from various vendors, manufacturers, and open-source projects—such as NVIDIA, OAI, OpenShift, Keysight, OSC, Open5GS, and Foxconn—thereby creating a truly multi-vendor network architecture. It currently comprises more than 8 RAN servers for the NVIDIA/OAI CU and DU (known as NVIDIA ARC-OTA and referred to as Aerial RAN CoLab (ARC) in this paper), several RUs from different vendors that can be installed in a lab space, as well as a Keysight RU emulator for further testing and profiling, O-RAN 7.2 fronthaul and timing hardware, along with multiple 5G CNs. The system delivers Key Performance Indicators (KPIs) representative of 5G sub-6 GHz systems, with cell throughput north of 1.65 Gbps with up to 25 connected UEs and a 100 MHz carrier bandwidth.

The tools we developed, integrated, and deployed on X5G can be readily used for the development of intelligent use cases for 5G and beyond, thanks to the combination of NVIDIA ARC, OAI, and the OSC projects. As a result, this combination offers performance improvements over most open-source, non-accelerated solutions while maintaining the openness and code accessibility typical of Open RAN systems, further enhanced by the seamless integration of GPUs. X5G provides researchers with the necessary capabilities to develop, test, and evaluate a wide range of AI/ML and RAN solutions on a production-ready platform, including spectrum sharing techniques [13], secure cellular networks [14], [15], resource optimization [16],

interference detection and mitigation, handover strategies, and the development of intelligent and autonomous networks. In addition, documentation and tutorials allow for the replication and bootstrapping of the testbed and its functionalities across research institutions and beyond. In fact, the value propositions of a platform similar to X5G, with its openness, multi-vendor support, and GPU-accelerated capabilities, have been demonstrated for industrial stakeholders, as shown by SoftBank and Fujitsu in [17].

The rest of the paper is organized as follows. Section II introduces the software frameworks we developed and integrated to enable X5G. Section III concerns the deployment and configuration of the X5G network infrastructure. Section IV describes an Radio Frequency (RF) planning study to determine an optimal location for deploying the RUs. System performance is evaluated in Section V through various use case scenarios with multiple Commercial Off-the-Shelf (COTS) UEs and applications. Section VI compares X5G with the state of the art. Section VII draws conclusions and outlines our future work.

## II. X5G SOFTWARE

This section describes the software components of X5G, also shown in Fig. 1. These components can be divided into three main groups: (i) a full-stack programmable Next Generation Node Base (gNB) (X5G RAN); (ii) the Open RAN RICs deployed on a micro-services cluster based on OpenShift; and (iii) various CNs deployed in a micro-services-based architecture essential for the effective functioning of the 5G network.

### A. Full-Stack Programmable RAN With NVIDIA Aerial and OpenAirInterface

The right part of Fig. 1 shows a detailed breakdown of the architecture of the X5G RAN, which follows the basic O-RAN architecture split into CU, DU, and RU. The DU is further split into a DU-low, implementing Layer 1 (PHY, or L1) functionalities, and into a DU-high, implementing Layer 2 (MAC and Radio Link Control (RLC), or L2) ones. As shown in Fig. 2, DU-low and DU-high communicate over the 5G FAPI interface specified by the SCF [18]. The DU-low is implemented using the NVIDIA Aerial Software Development Kit (SDK) [19] on in-line GPU accelerator cards, whereas DU-high and CU are implemented by OAI on general-purpose Central Processing Units (CPUs). We deploy each function in separate Docker containers, sharing a dedicated memory space for the inter-process communication library that enables the FAPI interface. In our setup, we also combine the CU and the DU-high into a combined L2/L3 gNB Docker container, but the F1 split has also been deployed and tested.

The FAPI interface between the DU-high and DU-low defines two sets of procedures: configuration and slot procedures. Configuration procedures handle the management of the PHY layer and happen infrequently, e.g., when the gNB stack is bootstrapped or reconfigured. On the contrary, slot procedures happen in every slot (i.e., every 500  $\mu$ s for a 30 kHz subcarrier spacing) and determine the structure of each Downlink (DL) and Uplink (UL) slot. In our case, L1 serves as the primary and L2 as

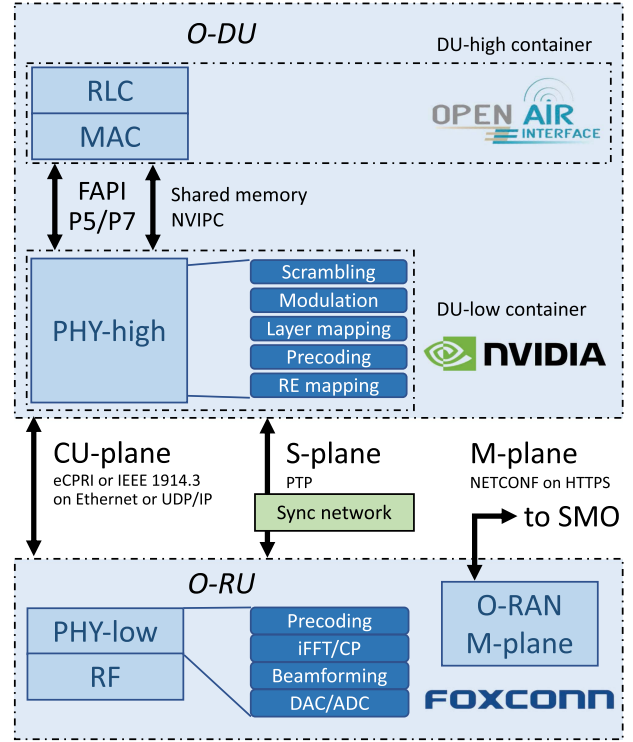


Fig. 2. Architecture of the lower layers of the X5G RAN following O-RAN specifications and consisting of: (i) a Foxconn O-RU; (ii) an O-DU-low based on NVIDIA Aerial SDK; (iii) an O-DU-high based on OpenAirInterface with their corresponding interfaces.

the subordinate. Upon the reception of a slot indication message from L1, L2 sends either an UL or DL request to dictate the required actions for the PHY layer in each slot. Additionally, L1 might transmit other indicators to L2, signaling the receipt of data related to Random Access Channel (RACH), Uplink Control Indication (UCI), Sounding Reference Signal (SRS), checksums, or user plane activities.

In our implementation, we use FAPI version 222.10.02 with a few exceptions as outlined in the NVIDIA Aerial release notes [20]. The transport mechanism for FAPI messages is specified in the networked FAPI (nFAPI) specification [21], which assumes that messages are transported over a network. However, in our implementation, the L1 and L2 Docker containers communicate through the NVIDIA Inter-Process Communication (NVIPC) library. This tool provides a robust shared memory communication framework specifically designed to meet the real-time performance demand of the data exchanges between MAC and PHY layers. In our implementation, we choose to transport the messages using little-endian with zero padding to 32 bits. The NVIPC library is also capable of tracing the FAPI messages and exporting them to a pcap file that can be analyzed offline with tools such as Wireshark.

The NVIDIA physical layer in the DU-low implements the O-RAN Open Fronthaul interface, also known as the O-RAN 7.2 interface [22], to communicate directly with the O-RU, in our case manufactured by Foxconn. This interface transports frequency domain In-phase and Quadrature (IQ) samples (with

TABLE I  
X5G ARC DEPLOYMENT MAIN FEATURES

| Feature                                                | Description               |
|--------------------------------------------------------|---------------------------|
| 3GPP Release                                           | 15                        |
| Frequency Band                                         | n78 (FR1, TDD)            |
| Carrier Frequency                                      | 3.75 GHz                  |
| Bandwidth ( $\beta$ )                                  | 100 MHz                   |
| Subcarrier spacing ( $\Delta f$ )                      | 30 kHz                    |
| Resource Block size ( $\chi$ )                         | 12 subcarriers            |
| Modulation order ( $Q_m$ )                             | 8 (256-QAM)               |
| TDD config                                             | DDDSU, DDDDDDSUUU*        |
| Number of antennas used                                | 4 TX, 4 RX                |
| MIMO config ( $L_{DL}, L_{UL}$ )                       | 4 layers DL, 1 layer UL   |
| Max theoretical cell throughput** ( $T_{DL}, T_{UL}$ ) | 1.64 Gbps DL, 204 Mbps UL |

\*Currently the special slot is unused due to limitations in Foxconn radios. \*\*The single-user maximum theoretical DL throughput can currently only be reached in the DDDDDDSUUU TDD configuration. In the DDDSU TDD configuration, it is limited to 350 Mbps since we can schedule a maximum of 2 DL slots per user in one TDD period, as only 2 ACK/NACK feedback bits are available per user.

optional block floating point compression) over a switched network, allowing for flexible deployments. The interface includes synchronization, control, and user planes. The synchronization plane, or S-plane, is based on PTPv2. We use synchronization architecture option 3 [23], where the fronthaul switch provides timing to both DU and RU. The interface also includes a management plane, although our system currently does not support it.

Table I summarizes the main features and operational parameters of the ARC deployment in the X5G testbed. The protocol stack is aligned with 3GPP Release 15 and uses the 5G n78 Time Division Duplexing (TDD) band and numerology 1. The DDDSU TDD pattern, which repeats every 2.5 ms, includes three downlink slots, one special slot (which is not used due to limitations in the Foxconn RUs), and an uplink slot. The uplink slot format implemented in OAI carries only two feedback bits for ACK/NACK per UE, thus allowing only the scheduling of two downlink slots per UE, eventually limiting the single UE throughput. Alternative TDD patterns, including DDDDDDSUUU and DDDDDDSUU, repeating every 5 ms, are also already in use to provide additional ACK/NACK bits for reporting from the UEs and mitigate this limitation.

To compute the maximum theoretical cell throughput in downlink ( $T_{DL}$ ) and uplink ( $T_{UL}$ ), we first derive a few additional parameters from Table I. The number of resource blocks ( $N_{RB}$ ) is computed using

$$N_{RB} = \frac{\beta}{\chi \cdot \Delta f} = 273. \quad (1)$$

By default, the number of Orthogonal Frequency Division Multiplexing (OFDM) symbols per slot ( $N_{sym}$ ) is 14. The number of slots per second ( $N_{slot}$ ) is inversely proportional to the slot duration, which for numerology  $\mu = 1$  is 0.5 ms. Hence,  $N_{slot} = 1s/0.5ms = 2000$  slots/second. The maximum theoretical cell throughput for downlink and uplink is given by

$$T_{DL,UL} = N_{RB} \cdot \chi \cdot N_{sym} \cdot N_{slot} \cdot Q_m \cdot R \cdot L_{DL,UL} \cdot \eta, \quad (2)$$

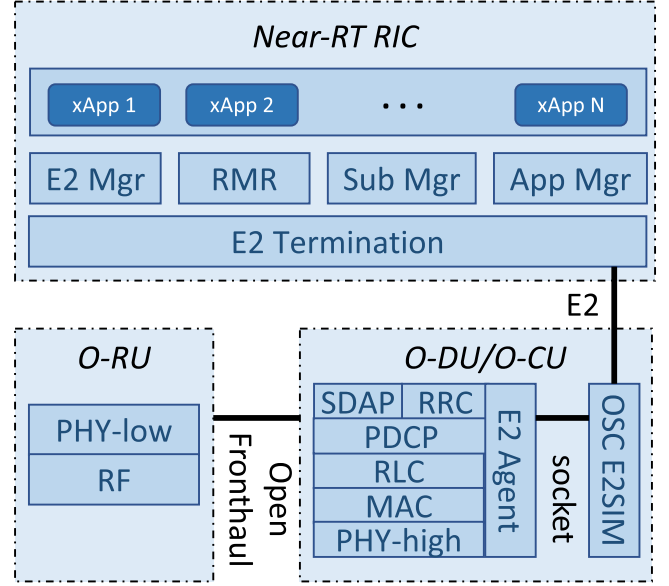


Fig. 3. Integration of the OSC Near-RT RIC in the OpenShift cluster with the X5G RAN.

where  $R$  is the effective code rate, which can approach 0.93 (as specified in the 3GPP standard [24]), and  $\eta$  is the fraction of time allocated for downlink or uplink operations based on the chosen TDD pattern. Considering the DDDDDDSUUU pattern to circumvent current OAI limitations on the ACK/NACK feedback bits, 60% of time is allocated for downlink and 30% for uplink since the special slot is unused due to Foxconn RU constraints. Consequently, the resulting theoretical peak cell throughput is 1.64 Gbps for downlink ( $T_{DL}$ ) and 204 Mbps for uplink ( $T_{UL}$ ). These values do not account for overheads typical of real networks—such as DeModulation Reference Signal (DMRS), Physical Uplink Control Channel (PUCCH), and Physical Downlink Control Channel (PDCCH)—which may further reduce net throughput. As shown by the experimental results in Section V-E, X5G peak performance nearly reaches the theoretical downlink throughput, while the uplink is still under improvement.

### B. Integration With the OSC Near-RT RIC

One of the key components of an O-RAN deployment is the Near-Real-Time (or Near-RT) RIC, and the intelligent applications hosted therein, namely xApps. These can implement closed-control loops at timescales between 10 ms and 1 s to provide optimization and monitoring of the RAN [25], [26]. In the current X5G setup, we deploy the “E” release of the OSC Near-RT RIC on a RedHat OpenShift cluster [4], which manages the lifecycle of edge-computing workloads instantiated as containerized applications. The Near-RT RIC and the ARC RAN are connected through the O-RAN E2 interface (see Fig. 3), based on Stream Control Transmission Protocol (SCTP) and an O-RAN-defined application protocol (E2AP) with multiple service models implementing the semantic of the interface (e.g., control, reporting, etc) [3]. On the gNB side, we integrate an E2

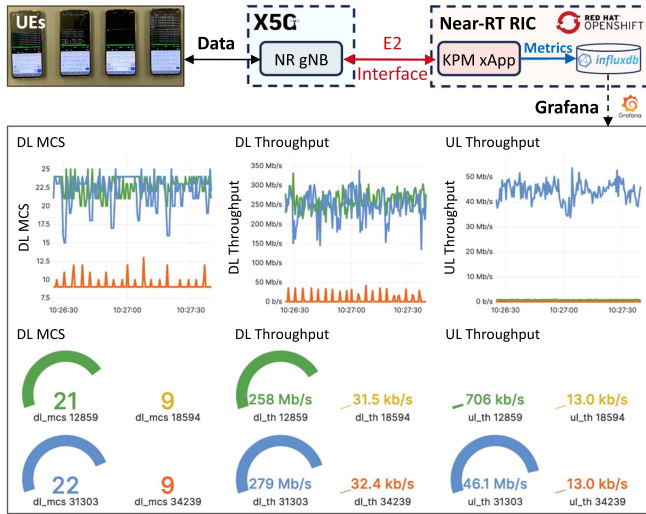


Fig. 4. KPM xApp example architecture including an X5G gNB with four connected UEs, each performing a different operation (ping, video streaming, DL test, and DL/UL tests), and a KPM xApp that pushes UE metrics into an Influx database, which are then visualized in a Grafana dashboard.

agent based on the *e2sim* software library [27], [28], which is used to transmit the metrics collected by the OAI gNB to the RIC via the Key Performance Measurement (KPM) E2 service model. These metrics are then processed by xApps deployed on the RIC, and used to compute some control action (e.g., through AI/ML agents) that is sent to the RAN through the E2 interface and processed by the *e2sim* agent.

As an example, Fig. 4 shows the architecture of a KPM xApp integrated with the X5G testbed. This xApp receives metrics from the E2 agent in the gNB, including throughput, number of UEs, and Reference Signal Received Power (RSRP), and stores them in an InfluxDB database [29]. The database is then queried to display the RAN performance on a Grafana dashboard [30] (see Fig. 4). This setup creates a user-friendly observation point for monitoring network performance and demonstrates the effective integration of the near-RT RIC in our configuration. A tutorial on how to deploy and run this xApp in X5G or on a similar testbed can be found on the OpenRAN Gym website [31], which hosts an open-source project and framework for collaborative research in the O-RAN ecosystem [32].

The metrics collected by a KPM xApp can then be leveraged by a second xApp or an rApp to perform smart closed-loop RAN controls at runtime, based on an arbitrary optimization strategy or specific requirements. ORANSlice—an open-source, network-slicing-enabled Open RAN system that leverages open-source RAN frameworks such as OAI [16]—was successfully integrated and tested in X5G, enabling near-real-time slicing control of the resources allocated by a gNB to multiple slices of the network, according to different policies set by the network manager. Fig. 5 presents the effects of various network policies applied by an ORANSlice slicing xApp in a X5G gNB. Fig. 5(a) shows the DL throughput results for the two slices (slice 1 in blue and slice 2 in orange), with a single UE per slice connected and transmitting 50 Mbps of DL User Datagram Protocol (UDP)

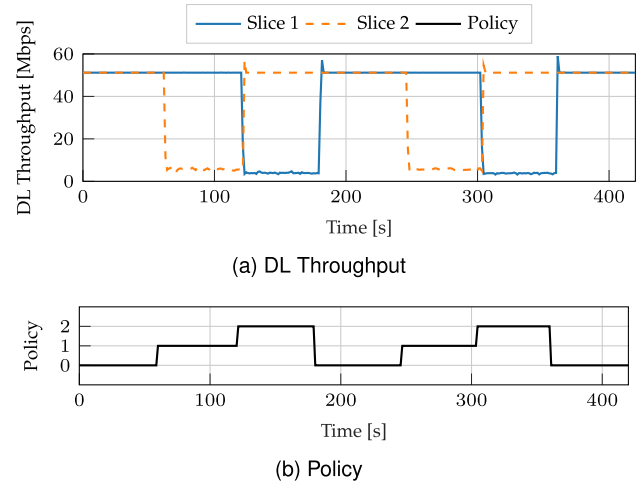


Fig. 5. Slicing xApp example showing: (a) DL throughput for two different slices, each with a single UE connected and pushing 50 Mbps of UDP traffic; (b) the network policy applied by the slicing xApp, switching between no-priority (0), prioritize slice 1 (1), and prioritize slice 2 (2).

data, according to the policy shown in Fig. 5(b). The slicing xApp switches between three policies: (0) no-priority, where all slices share all resources, so both UEs achieve the target throughput of 50 Mbps; (1) prioritize slice 1: where 98% of resources are reserved for the first slice and 2% for the second one, causing the latter performance to drop to only 6 Mbps; (2) prioritize slice 2, where the opposite behavior of policy 1 is observed, with slice 1 now unable to achieve the target throughput. In this example, the policy is applied arbitrarily as a proof-of-concept for the network slicing control capabilities of X5G, while more intelligent strategies employing AI/ML components can be easily integrated into the decision process. Additional applications, including the emerging dApps [13], are currently being integrated into X5G to fully leverage its openness and programmability, further demonstrating the benefits of smart closed-loop control within the O-RAN ecosystem.

### C. Core Network

The X5G testbed facilitates the integration and testing of different CNs from various vendors and projects. We leverage virtualization to deploy all the necessary micro-services, e.g., Access and Mobility Management Function (AMF), Session Management Function (SMF), User Plane Function (UPF), in the OpenShift cluster that also supports the Near-RT RIC. We have successfully tested and integrated the X5G RAN with two open-source core network implementations, i.e., the 5G CNs from OAI [12], as also discussed in [1], and, in this paper, also with Open5GS [33] and the CoreSIM software from Keysight [34]. As part of our ongoing efforts, we plan to incorporate additional cores, including the commercial core from A5G [35].

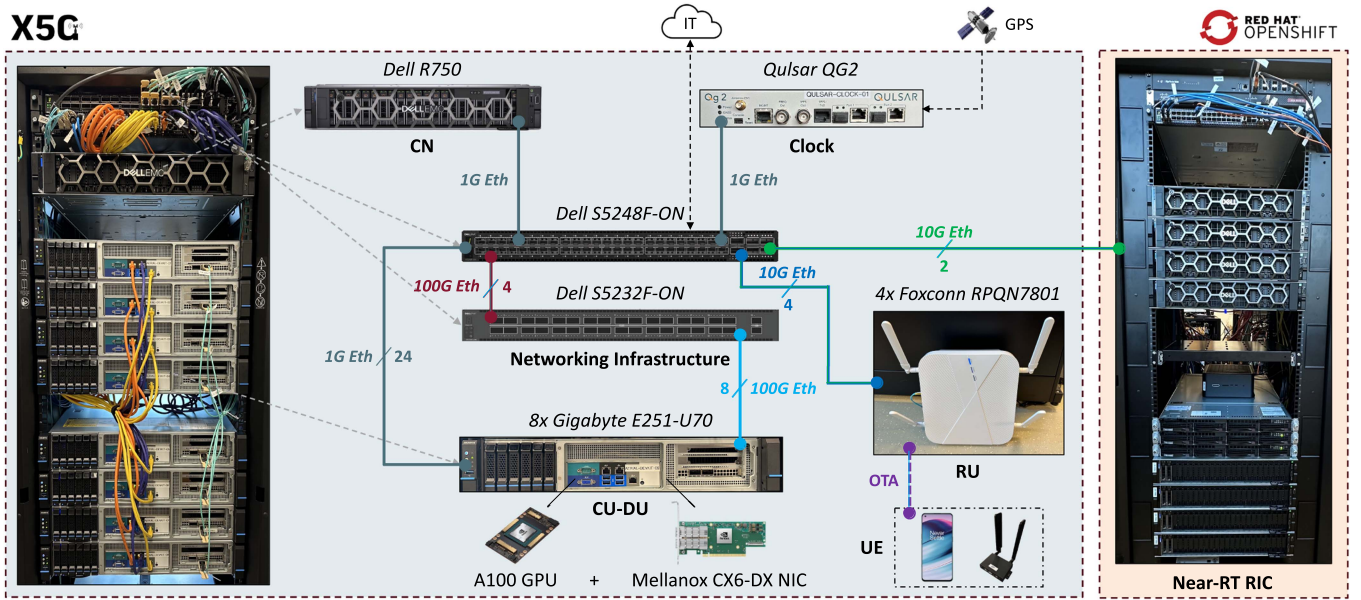


Fig. 6. Hardware and architecture infrastructure of the X5G deployment at Northeastern University.

#### D. X5G Software Licensing and Tutorials

X5G, including the Aerial PHY, the OAI higher layers, as well as the OSC RIC, is open and can be extended with custom features and functionalities. The NVIDIA ARC framework is documented on the NVIDIA portal [20], which is accessible through NVIDIA's 6G developer program. As mentioned in Section II-B, the step-by-step integration between the OSC RIC and the ARC stack through the X5G E2 agent is discussed in a tutorial on the OpenRAN Gym website [31], [32].

The components implemented by OAI are published under the OAI public license v1.1 created by the OAI Software Alliance (OSA) in 2017 [36]. This license is a modified Apache v2.0 License, with an additional clause that allows contributors to make patent licenses available to third parties under Fair, Reasonable, And Non-Discriminatory (FRAND) terms, similar to 3GPP for commercial exploitation, to allow contributions from companies holding intellectual property in related areas. The usage of OAI code is free for non-commercial/academic research purposes. The Aerial SDK is available through an early adopter program [20]. The OSC software is published under the Apache v2.0 License.

### III. X5G INFRASTRUCTURE

This section describes the X5G physical deployment that is currently located on the Northeastern University campus in Boston, MA.<sup>1</sup> The deployment includes a server room with a dedicated rack for the private 5G system and an indoor laboratory open space area with benches and experimental equipment that provide a realistic RF environment with rich scattering and obstacles. Fig. 6 illustrates the hardware infrastructure that we

deployed to support the X5G operations. This includes synchronization and networking infrastructures, radio nodes, eight ARC servers with integrated DU and CU, and additional compute infrastructure for the RIC and CN deployments. This infrastructure, which will be described next, has been leveraged to provide connectivity for up to eight concurrent COTS UEs, such as OnePlus smartphones (AC Nord 2003) and Sierra Wireless boards (EM9191) [37].

**Synchronization Infrastructure:** The synchronization infrastructure consists of a Qulsar (now VIAVI) QG-2 device acting as grandmaster clock. The QG-2 unit is connected to a GPS antenna for precise class-6 timing and generates both Precision Timing Protocol (PTP) and Synchronous Ethernet (SyncE) signals to provide frequency, phase, and time synchronization compliant with the ITU-T G.8265.1, G.8275.1, and G.8275.2 profiles. It sends the synchronization packets to the networking infrastructure through a 1 Gbps Ethernet connection. The networking infrastructure then offers full on-path support, which is necessary to distribute phase synchronization throughout the X5G platform.

**Networking Infrastructure:** The networking infrastructure provides connectivity between all the components of the X5G platform. It features fronthaul and backhaul capabilities through the use of two Dell switches (S5248F-ON and S5232F-ON) interconnected via four 100 Gbps cables in a port channel configuration. This configuration allows for the aggregation of multiple physical links into a single logical one to increase bandwidth and provide redundancy in case some of them fail. All switch ports are sliced into different Virtual Local Area Networks (VLANs) to allow the proper coexistence of the various types of traffic (i.e., fronthaul, backhaul, management). The Dell S5248F-ON switch primarily provides backhaul capabilities to the network and acts as a boundary clock in the synchronization plane, receiving PTP signals from the synchronization infrastructure. This switch

<sup>1</sup>X5G website: <https://x5g.org>.

includes 48 SFP+ ports: 12 ports are dedicated to the RUs and receive PTP synchronization packets, 10 are used to connect to the OpenShift cluster and service network, 10 are used for the out-of-band management network, and 16 connect to the CN and the Internet. Additionally, the switch includes 6 QSFP28 ports, 4 of which interconnect with the second switch. The Dell S5232F-ON switch mainly provides fronthaul connectivity to the gNBs. It includes 32 QSFP28 ports: 8 ports connect to the Mellanox cards of the ARC nodes via 100 Gbps fiber links, and 4 connect to the Dell S5248F-ON switch. The latter also acts as a boundary clock, receiving the synchronization messages from the S5232F-ON and delivering them to the gNBs.

**RU:** We deployed eight Foxconn RPQN 4T4R RUs, operating in the n78 band, with additional units being tested in the lab, and the Keysight RuSIM emulator. The Foxconn units have 4 externally mounted antennas, each antenna with a 5 dBi gain, and 24 dBm of transmit power. The Over-the-Air (OTA) transmissions are regulated as part of the Northeastern University Federal Communications Commission (FCC) Innovation Zone [38], with an additional transmit attenuation of 20 dB per port to comply with transmit power limits and guarantee the coexistence of multiple in-band RUs in the same environment. As we will discuss in Section V, we leverage two of these RUs for the experimental analysis presented in this work. These RUs are deployed following RF planning procedures discussed in Section IV. Plans are in place to procure Citizen Broadband Radio Service (CBRS) RUs and deploy them in outdoor locations. RUs from additional vendors are being tested and integrated as part of our future works. Finally, we also tested and integrated the ARC stack with the Keysight RuSIM emulator, which supports the termination of the fronthaul interface on the RU side and exposes multiple RUs to the RAN stack, for troubleshooting, conformance testing, and performance testing [39].

**CU and DU:** The 8 ARC nodes that execute the containerized CU/DU workloads are deployed on Gigabyte E251-U70 servers with 24-core Intel Xeon Gold 6240R CPU and 96 GB of RAM. The servers—which come in a half rack chassis for deployment in RAN and edge scenarios—are equipped with a Broadcom PEX 8747 Peripheral Component Interconnect (PCI) switch that enables direct connectivity between cards installed in two dedicated PCI slots without the need for interactions with the CPU. Specifically, the two PCI slots host an NVIDIA A100 GPU, which supports the computational operations of the NVIDIA Aerial PHY layer, as well as a Mellanox ConnectX-6 Dx Network Interface Card (NIC). The latter, which is used for the fronthaul interface, connects to the fronthaul part of the networking infrastructure via a QSFP28 port and 100 Gbps fiber-optic cable. In this way, the NIC can offload or receive packets directly from the GPU, thus enabling low-latency packet processing. Finally, each server is connected to the backhaul part of the networking infrastructure through three 1 Gbps Ethernet links, which provide connectivity with the OpenShift cluster (and thus the Near-RT RIC and the core networks), the management infrastructure, and the Internet. In addition to the Gigabyte servers, seven Grace Hopper (GH) machines—one of the latest NVIDIA high-computing ARM-based devices—are currently

being integrated into X5G to run the ARC CU/DU workloads. Each GH combines a 72-core NVIDIA Grace CPU Super-chip and an NVIDIA H100 Tensor Core GPU, linked through NVIDIA NVLink-C2C technology, which ensures seamless data sharing with up to 900 GB/s of bandwidth. It also features 480 GB of RAM, two BlueField-3 Data Processing Units (DPUs), and ConnectX-7 NICs. This configuration provides significantly higher computational capabilities compared to the Gigabyte servers, enabling the efficient support of concurrent RAN and AI/ML workloads.

*Additional Compute:* We leverage additional servers that are part of the OpenShift cluster and are used to instantiate the various CNs and the Near-RT RIC. The OpenShift cluster includes three Dell R740 servers acting as control-plane nodes and two Microway Navion Dual servers as worker nodes. The OpenShift rack is linked to the X5G rack through two 10 Gbps connections, one dedicated to OpenShift operations, and the other for the out-of-band management. Additionally, a Dell R750 server with 56 cores and 256 GB RAM is available for the deployment and testing of additional core network elements. This server connects to the networking infrastructure via a 1 Gbps Ethernet link and has access to the Internet through the Northeastern University network.

#### IV. RF PLANNING WITH RAY-TRACING

In this section, we present RF planning procedures to identify suitable locations for the RU deployment. This approach leverages an exhaustive search within a ray-tracing-based digital twin framework, with the objective of maximizing the RUs coverage while minimizing the overall interference. The study is conducted only once during the system deployment phase and remains valid as long as no significant changes occur in the environment. We perform ray-tracing in a detailed digitized representation of our indoor laboratory space in the Northeastern University ISEC building in Boston, MA, to achieve high fidelity between the real-world environment and the digital one. We carefully study how to deploy 2 RUs by considering the Signal to Interference plus Noise Ratio (SINR) between the RUs and the UEs as the objective function in the optimization problem. We limit the optimization space by using a grid of 24 possible RU locations and 52 UE test points, enabling an exhaustive search approach instead of a formal integer optimization problem, since these constraints keep the computation manageable. First, we leverage the 3D representation of our laboratory space, created as part of the digital twin framework developed in [40] using the SketchUp modeling software. We then import the model in the MATLAB ray-tracing software and define the locations of RUs and UEs as shown in Fig. 7(a) (from a top perspective) and in Fig. 7(b) (from a side view). The 24 possible RUs locations (2 for each bench) are shown in red, while the 52 test points for the UEs (arranged in a  $4 \times 13$  grid) are in blue.

Tables I and II summarize the parameters used in our ray-tracing model. For the deployment planning purpose, we consider the RUs as transmitter nodes (TX) and the UEs as receiver ones (RX), i.e., we tailor our deployment to downlink transmissions.

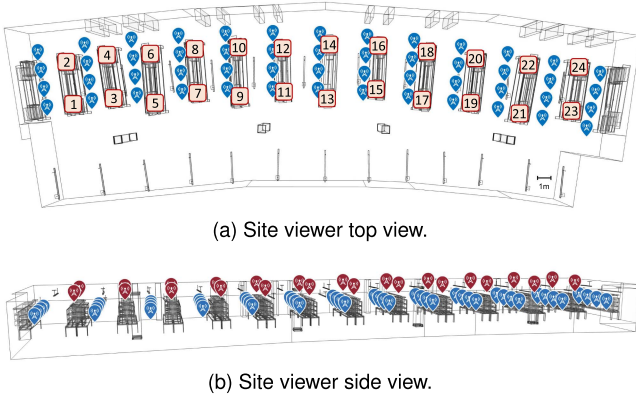


Fig. 7. Site viewer with RU (red squares) and UE (blue icons) locations.

TABLE II  
PARAMETERS OF THE MATLAB RAY-TRACING STUDY TO DETERMINE RU LOCATIONS

| Parameter                              | Value                      |
|----------------------------------------|----------------------------|
| RU antenna spacing                     | 0.25 m                     |
| RU antenna TX power ( $P_{RU}$ )       | 24 dBm                     |
| RU antenna gain ( $G_{RU}$ )           | 5 dBi                      |
| RU antenna pattern                     | Isotropic                  |
| RU TX attenuation ( $A_{RU}$ )         | [0 – 50] dB                |
| Set of RU locations ( $\mathcal{R}$ )  | 24 in a $2 \times 12$ grid |
| RU height                              | 2.2 m                      |
| UE number of antennas                  | 2                          |
| UE antenna spacing                     | 0.07 m                     |
| UE antenna gain ( $G_{UE}$ )           | 1.1 dBi                    |
| UE noise figure ( $F_{UE}$ )           | 5 dB                       |
| Set of UEs locations ( $\mathcal{U}$ ) | 52 in a $4 \times 13$ grid |
| UE height                              | 0.8 m                      |
| Environment material                   | Wood                       |
| Max number of reflections              | 3                          |
| Max diffraction order                  | 1                          |
| Ray-tracing method                     | Shooting and bouncing rays |

The ray-tracer generates a  $24 \times 52$  matrix  $\mathbf{C}$  where each entry  $c_{i,j}$  corresponds to the channel information between  $RU_i$  with  $i \in \mathcal{R}, \mathcal{R} = 1, \dots, 24$ , and  $UE_j$  with  $j \in \mathcal{U}, \mathcal{U} = 1, \dots, 52$ . We use this to derive relevant parameters such as the thermal noise ( $\mathcal{N}$ ) and the path loss ( $\mathcal{L}$ ) to compute the Received Signal Strength Indicator (RSSI)  $\mathcal{S}_{i,j}$  for  $UE_j$  connected to  $RU_i$ , as follows:

$$\mathcal{S}_{i,j} = P_{RU,i} + G_{RU,i} - A_{RU,i} - \mathcal{L}_{i,j} + G_{UE,j}, \quad (3)$$

where  $P_{RU,i}$ ,  $G_{RU,i}$ , and  $A_{RU,i}$  are the antenna TX power, gain, and attenuation of  $RU_i$ , respectively. Then, considering the linear representation of  $\hat{\mathcal{S}}_{i,j}$ , the SINR  $\Gamma_{i,j}$  is

$$\Gamma_{i,j} = \frac{\hat{\mathcal{S}}_{i,j}}{\mathcal{N}F_{UE,i} + \sum_{u=1, u \neq i}^M \hat{\mathcal{S}}_{u,j}}, \quad (4)$$

where  $M$  is the number of RUs being deployed,  $\mathcal{N}$  is the thermal noise, and  $F_{UE,i}$  is the noise figure of  $UE_i$ . The SINR  $\Gamma_{i,j}$  considers the interference to the signal from  $RU_i$  to  $UE_j$  due to downlink transmissions of all other  $M - 1$  RUs being deployed.

In our RF planning, we deploy two RUs (i.e.,  $M = 2$ ). In the following study, we consider scenarios where the first RU

### Algorithm 1: Exhaustive Search Algorithm for RF Planning.

**Input:** Set of RU locations ( $\mathcal{R}$ ), set of UE test points ( $\mathcal{U}$ ), precomputed SINR matrix  $\Gamma$   
**Output:** Optimal RU pair ( $p^*, q^*$ ) and maximum average SINR  $\Phi_{\max}(\Gamma)$

- 1: Initialize  $bestScore \leftarrow -\infty$  and  $bestPair \leftarrow \text{None}$
- 2: **for all** RU pairs  $(p, q) \in \binom{\mathcal{R}}{2}$  **do**
- 3:    $sumSINR \leftarrow 0$
- 4:   **for all** UE  $j \in \mathcal{U}$  **do**
- 5:     Compute  $\Gamma_{p,j}$  (RU  $p$  serving, RU  $q$  interfering)
- 6:     Compute  $\Gamma_{q,j}$  (RU  $q$  serving, RU  $p$  interfering)
- 7:      $sinarMax \leftarrow \max(\Gamma_{p,j}, \Gamma_{q,j})$
- 8:      $sumSINR \leftarrow sumSINR + sinarMax$
- 9:   **end for**
- 10:    $avgSINR \leftarrow sumSINR/|\mathcal{U}|$
- 11:   **if**  $avgSINR > bestScore$  **then**
- 12:      $bestScore \leftarrow avgSINR$
- 13:      $bestPair \leftarrow (p, q)$
- 14:   **end if**
- 15: **end for**
- 16: **return**  $bestPair, bestScore$

TABLE III  
BEST RUS AND AVERAGE SINR  $\Phi(\Gamma)$  RANGE VALUES

| $A_{RU}$ [dB] | RU locations with best SINR | [Min, Max] $\Phi(\Gamma)$ [dB] |
|---------------|-----------------------------|--------------------------------|
| 0             | [8, 23]                     | [6.08, 23.33]                  |
| 10            | [6, 23]                     | [5.71, 22.66]                  |
| 20            | [6, 23]                     | [5.00, 21.03]                  |
| 30            | [8, 23]                     | [3.58, 17.82]                  |
| 40            | [7, 24]                     | [0.19, 12.94]                  |
| 50            | [8, 20]                     | [-6.23, 6.63]                  |

serves one UE from the test locations, while we assume that the second RU creates interference with the first, even without being assigned any UE from the list. We test all possible combinations of the 24 RU test locations, which, following the combinatorial equation of choosing 24 elements ( $n$ ) in groups of 2 ( $r$ ) as  $C(n, r) = \frac{n!}{r!(n-r)!}$ , results in a total of 276 pairs. The proposed approach for determining the optimal RU locations and the maximum average SINR ( $\Phi_{\max}(\Gamma)$ ), called score, is presented in Algorithm 1. It takes as input the set of RU locations ( $\mathcal{R}$ ), the set of UE test points ( $\mathcal{U}$ ), and the SINR matrix  $\Gamma$ . Then, it performs an exhaustive search, testing all pairs of RU against all UEs to determine the optimal RU pair ( $p^*, q^*$ ) with the best maximum average SINR  $\Phi_{\max}(\Gamma)$ .

We test this algorithm with different values of the attenuation  $A_{RU}$ , from 0 to 50 dB in 10 dB increments.

Fig. 8 visualizes the normalized values of the score  $\Phi(\Gamma)$  for all possible combinations of RU pairs and different attenuation values. Additionally, Table III provides the best RU locations including the minimum and maximum values of  $\Phi(\Gamma)$  for the corresponding combinations. As expected, locations with further RUs exhibit higher average SINR values, as they are less affected by interference. However, it is important to note that

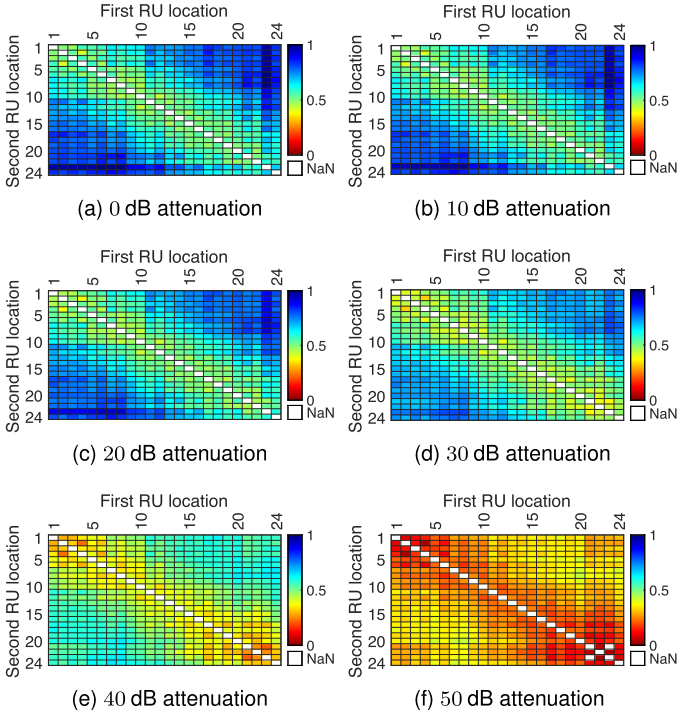


Fig. 8. Heatmap results of the normalized average SINR  $\Phi(\Gamma)$  with 2 RUs.

the score also considers coverage, as it is computed based on the SINR. Consequently, the optimal combination of locations identifies RUs that are further apart but not necessarily the furthestmost pair. Considering these results, for the experiments in Section V, we select a TX attenuation of 20 dB, which exhibits a good trade-off between coverage and average SINR values. Moreover, during our real-world experiments, we observed that a 20 dB attenuation leads to increased system stability and reduced degradation compared to lower attenuation values, resulting in improved overall performance, as it reduces the likelihood of saturation at the UE antenna. Therefore, we select locations [6,23] for our RUs deployment.

## V. EXPERIMENT RESULTS

In this section, we describe the design and execution of a comprehensive set of experiments that illustrate the capabilities of the X5G infrastructure in a variety of operational scenarios. We assess the adaptability of the testbed through rigorous testing, utilizing iPerf to measure network throughput and MPEG-DASH to gauge video streaming quality. The experiments are mainly conducted in the same indoor laboratory area modeled in Section IV. They include static configurations with a single UE as well as more complex setups with multiple UEs and RUs, leveraging the Keysight RuSIM emulator, and scenarios with UE mobility.

### A. Setup Overview

We consider two different setups: (i) Gigabyte RAN servers with a 2x2 Multiple Input, Multiple Output (MIMO) configuration ( $L_{DL}, L_{UL}$ ), 2 layers DL, 1 layer UL, a DDDSU TDD pattern, and a modulation order ( $Q_m$ ) up to 64-QAM (results

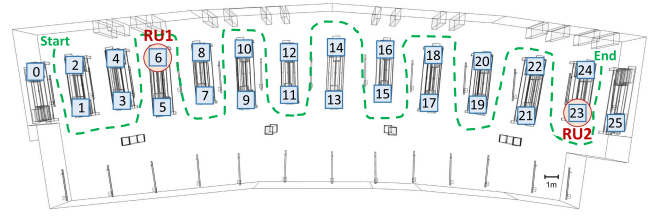


Fig. 9. Node locations considered in our experiments: RUs (red circles in 6 and 23); possible static UEs (blue squares); and mobile UEs (green dashed line).

for this setup are shown in Sections V-B, V-C, and V-D); and (ii) GH RAN servers with a 4x4 MIMO configuration, 4 layers DL, 1 layer UL, a DDDDDSUUU TDD pattern, and a  $Q_m$  up to 256-QAM (Sections V-E and V-F). All experiments utilize a carrier frequency of 3.75 GHz with a bandwidth ( $\beta$ ) of 100 MHz.

The experiments are mainly conducted in the laboratory area shown in Fig. 9, which highlights the RU locations (outcome of the ray-tracing study discussed in Section IV), as well as the UE locations and the mobility pattern for the non-static experiments. All tests involving a single RU are conducted at location 6, as illustrated in Fig. 9. An edge server, configured to support the iPerf and MPEG-DASH applications, is deployed within the campus network to ensure minimal latency, ranging from 1 to 2 ms. During static throughput tests, Transmission Control Protocol (TCP) backlogged traffic is transmitted first in the downlink and then in the uplink directions for 40 seconds each across different UE configurations. For video streaming, the server employs FFmpeg [41] to deliver five distinct profiles simultaneously at various resolutions—ranging from 1080P at 250 Mbps to 540P at 10 Mbps—to the UEs. On the device side, we leverage a pre-compiled iPerf3 binary for Android to generate TCP traffic, and Google's ExoPlayer for client-side video playback. Each set of experiments is replicated five times to ensure data reliability, with results including mean values and 95% confidence intervals of the metrics plotted. These metrics encompass application layer measurements such as throughput, bitrate, and rebuffer ratio, alongside MAC layer metrics like SINR, RSRP, and Modulation and Coding Scheme (MCS), collected at the OAI gNB level.

### B. Static Experiments

*1 UE, static, iPerf:* In the initial series of tests, we analyze the performance of a single UE in ten static locations at varying distances from the RU, as shown in Fig. 10, using the first configuration setup of 2x2 MIMO, 2 layers DL, 1 layer UL, a DDDSU TDD pattern, and a  $Q_m$  up to 64-QAM.

The iPerf throughput results in Fig. 10(a) highlight the upper layer's responsiveness, showing a significant reduction from an average downlink throughput of 300 Mbps and an uplink one of 38 Mbps at locations near the RU, to significantly lower rates of 177 Mbps in downlink and 1.5 Mbps in uplink at the most remote point, i.e., location 18. This high-level data throughput behavior is supported by corresponding shifts in the lower layers.

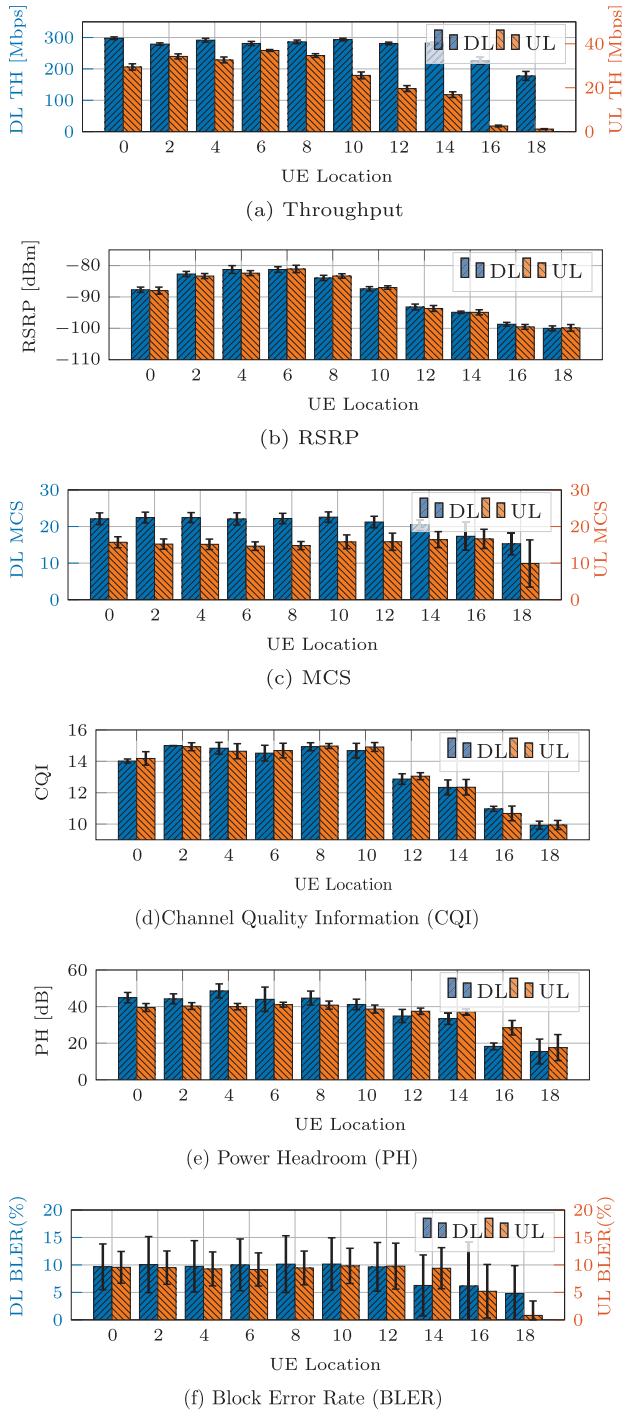


Fig. 10. Performance profiling with one UE and single RU for the static iPerf use case during DL (blue bars) and UL (orange bars) data transmissions.

For example, this trend is clearly noticeable from the results of Fig. 10(b), which shows the RSRP values reported by the UE to the gNB during DL (blue bars) and UL (orange bars) data transmissions. As the distance from the RU initially decreases starting from location 0 to location 6, the RSRP values peak at around  $-80$  dBm. Subsequently, as the distance starts to increase again, moving from location 6 towards location 18, the RSRP values begin to decrease, reaching as low as  $-100$  dBm.

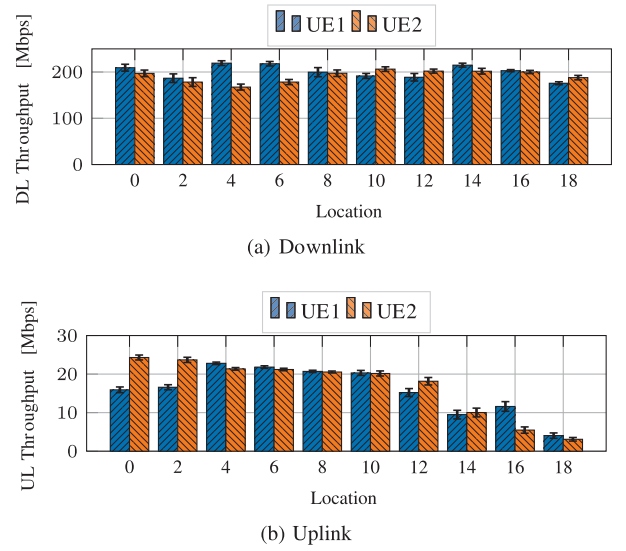


Fig. 11. Performance profiling for one RU and two static UEs for the static iPerf use case.

In the same way, the MCS values for both downlink and uplink initially mirror each other, then begin to diverge from around location 10, as shown in Fig. 10(c). In uplink, the MCS values tend to remain stable or slightly increase, whereas downlink MCS values experience a slight decline possibly due to power control strategies that better favor the uplink at more distant locations.

Similarly, the Channel Quality Information (CQI) (Fig. 10(d)) for both downlink and uplink starts closely matched but begins to show variation past the midpoint of the locations. The downlink CQI experiences a modest decline, while the uplink CQI sustains higher values. This suggests better channel conditions or more effective adaptation mechanisms in the uplink, due to adaptive power adjustments in uplink transmissions that maintain signal quality over distance.

PH metrics, shown in Fig. 10(e), reveal that downlink power headroom remains relatively stable across all locations, indicating a consistent application of power levels for downlink transmissions. In contrast, the uplink displays greater variability and generally higher values in distant locations to compensate for potential path loss and ensure that the transmit power remains adequate to maintain quality of service as the UE moves further from the RU.

Finally, the Block Error Rate (BLER) (Fig. 10(f)) for downlink remains below 10% for most locations, pointing to good reliability and effective adaptation of the MCS. However, while the uplink BLER is generally low, it exhibits some peaks, especially around mid-range locations which may be related to specific lab obstacles or multipath effects and slow adaptation loops.

**2 UEs, static, iPerf:** We test the performance of 2 UEs for a single RU. We position the UEs at the same static locations as in the previous single UE static case. The results are plotted in Fig. 11 for both downlink and uplink transmissions. We observe that, in most cases, the UEs are able to share bandwidth fairly.

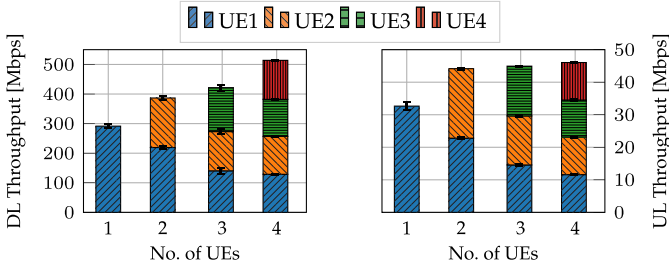


Fig. 12. Performance profiling with multiple UEs at fixed location 4 for the static iPerf use case using a DDDSU TDD pattern, a 2x2 MIMO configuration, 2 layers DL and 1 layer UL.

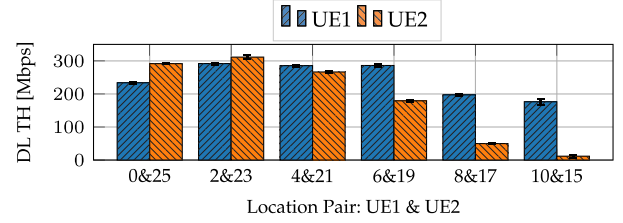
The best achievable average aggregate throughput from both UEs is around 400 Mbps in DL and 44 Mbps in UL. This shows that the total cell throughput can be higher than the single UE throughput. As discussed in Table I, this is due to a limitation in the number of transport blocks that can be acknowledged in a single slot for a single UE in the case of the DDDSU TDD pattern used in this first set of experiments. Therefore, scheduling multiple UEs improves the resource utilization of the system.

*1–4 UEs, static, iPerf:* We further extend our evaluation to include additional tests with multiple UEs. At fixed location 4, we compare system performance with varying numbers of UEs (from 1 to 4) connected to our network. The average throughput and 95% confidence intervals are plotted in Fig. 12. We observe that the UEs achieve steady throughput in all the cases, as indicated by the small confidence interval values. Additionally, the combined throughput increases with the number of UEs connected: with four UEs, the aggregate throughput reaches 512 Mbps in DL and 46 Mbps in UL. This scenario highlights the maximum throughput performance that X5G is able to achieve with the current  $2 \times 2$  MIMO configuration, featuring 2 layers in DL and a DDDSU TDD pattern, ensuring a fair distribution of resources among all UEs. It is worth noting that the peak performance of X5G is detailed in Section V-E.

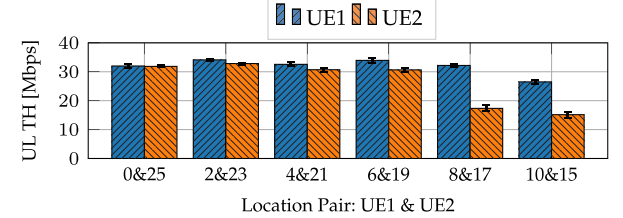
*2 RUs, 1 UE per RU, static, iPerf:* Finally, we evaluate X5G performance with two RUs by connecting UE1 to RU1 and UE2 to RU2. RU1 is located at position 6, and RU2 is at position 23. We select six pairs of locations—(0,25), (2,23), (4,21), (6,19), (8,17), (10,15)—for the UEs to ensure different distances among them and the RU.

From Fig. 13(a), we observe that the DL throughput is significantly impacted by interference, particularly at cell edge locations. The throughput for UE1 shows a reduction of up to 90% as the UEs approach each other, while UE2 throughput decreases by up to 50%. These observations indicate that interference predominantly affects the DL direction. Conversely, as depicted in Fig. 13(b), UL throughput remains relatively stable across different location pairs, suggesting that UL is less susceptible to the types of interference affecting DL throughput.

Fig. 14 further supports these observations by presenting additional KPIs from the MAC layer. Fig. 14(a) shows that the MCS for both UEs decreases as the distance between the UEs diminishes, indicative of increasing interference levels.

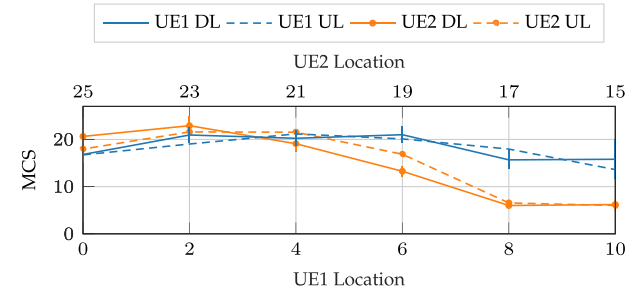


(a) Downlink

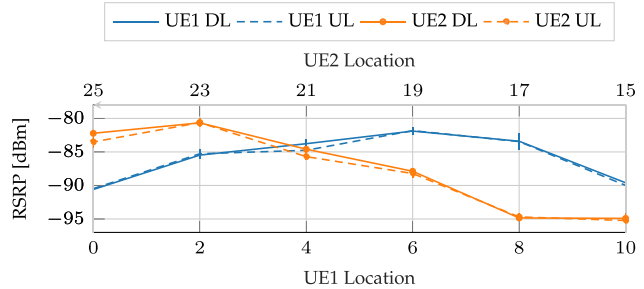


(b) Uplink

Fig. 13. Performance profiling for two RUs in the static iPerf use case, each with one assigned UE: UE1 to RU1 and UE2 to RU2.



(a) MCS



(b) RSRP

Fig. 14. MAC KPIs in the two RUs iPerf use case, each with one static UE (UE1 assigned to RU1 and UE2 to RU2): (a) averages and confidence intervals for DL MCS (solid lines) during DL data transmissions, and UL MCS (dashed lines) during UL transmissions, for UE1 (blue) and UE2 (orange); (b) averages and confidence intervals of RSRP reported by UE1 (blue) and UE2 (orange) during DL (solid lines) and UL (dashed lines) transmissions.

Fig. 14(b) illustrates the RSRP, which varies in response to the UEs locations. Notably, despite adequate RSRP levels, the throughput remains low, highlighting the significant impact of interference, particularly in the DL direction.

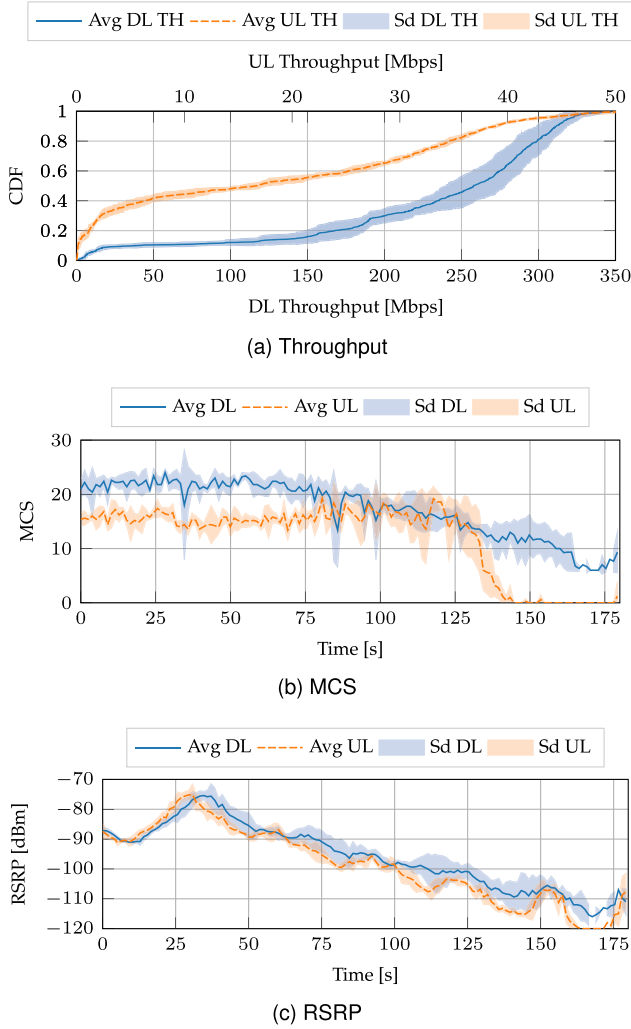


Fig. 15. Performance profiling with one RU and one mobile UE in the iPerf use case: (a) CDF of DL and UL throughputs with averages (solid lines) and SD (shaded areas); (b) averages (solid lines) and SD (shaded areas) of the DL MCS during DL transmissions (blue) and of the UL MCS during UL transmissions (orange); (c) averages (solid lines) and SD (shaded areas) of the RSRP reported by the UE during DL (blue) and UL (orange) data transmissions.

### C. Mobile Experiments

We assess the network performance by measuring throughput as the UE follows the walking pattern around the laboratory space depicted by the dashed green line in Fig. 9. The entire walk from the start to the end point spans approximately 3 minutes at regular walking speed. The mobile use case results are illustrated in Fig. 15. The application layer throughput is depicted by Cumulative Distribution Function (CDF) plots in Fig. 15(a), where solid lines represent the averaged curve for all runs, while the shaded areas around these lines illustrate the variation across different runs, indicating the range of values within one Standard Deviation (SD) above and below the mean. The MCS and RSRP results at the MAC layer are shown in Fig. 15(b) and (c), respectively. Also here, the average values are depicted using solid lines, while the shaded areas indicate the SD.

The throughput results of Fig. 15(a) highlight notable variability in network quality influenced by mobility. Throughout the test, the UE achieves peaks of up to 350 Mbps in DL

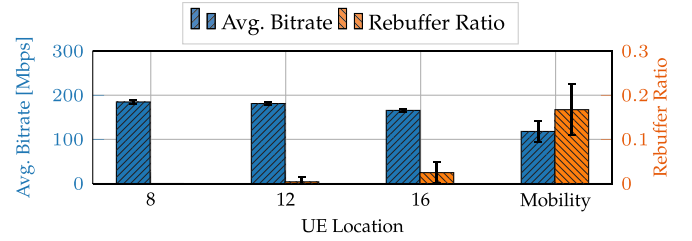


Fig. 16. Video streaming performance with one UE and single RU across both static (8—close, 12—mid, 16—far) and mobile use cases.

and 50 Mbps in UL. However, significant fluctuations in performance are observed, particularly as the UE moves further from the initial RU position. Fig. 15(b) illustrates a significant drop in UL MCS values around the 100-second mark, where averages initially above 10 drop sharply, while DL MCS fluctuate more gradually until they fall below 10. This sudden decline in UL MCS at this specific time is likely due to deteriorating signal conditions, as corroborated by the corresponding RSRP trends in Fig. 15(c). This is most probably due to increased distance from the base station or physical obstructions, leading to a necessary reduction in MCS to maintain connectivity under compromised signal strength.

The MCS results of Fig. 15(b) show that both DL and UL MCS values start relatively high but decrease as the UE moves further from the RU. The DL MCS exhibits more variability and sharper declines compared to the UL, which maintains a more stable profile until the final part of the walk. This suggests that the uplink benefits from more aggressive modulation and coding strategies due to 5G adaptive power control mechanisms that mitigate the impact of increasing distance and obstacles more effectively. Additionally, the RSRP data, shown in Fig. 15(c), indicates a gradual decline in signal strength as the UE moves along its trajectory. RSRP values for both DL (blue) and UL (orange) cases decrease over time, with the most significant drops observed after 100 seconds. This reduction in signal quality corresponds with declines in throughput and MCS, highlighting the strong dependency of these metrics on signal strength.

### D. Video Streaming Experiments

We place the UE at three static locations at different distances from the RU: location 8 (close); 12 (mid); and 16 (far). We run each video session for three minutes, streaming five distinct profiles simultaneously at various resolutions as described in Section V-A. We then plot the mean bitrate over five runs, as well as the rebuffer ratio, in Fig. 16. As expected, the average bitrate decreases and the rebuffer ratio increases as further distances between UE and RU are considered, transitioning from close to far static locations. We observe that the UE can achieve a steady mean bitrate of around 180 Mbps in all static cases. Note that, unlike test results achieved through iPerf backlogged traffic, the mean bitrate for video streaming is lower. The video client fetches segments in an intermittent fashion (causing flows to be short), which depends on parameters, e.g., video buffer and segment size. Because of this, throughput sometimes does not increase to the fullest during that short period of time,

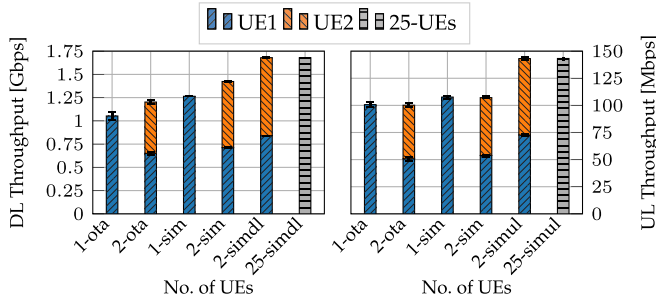


Fig. 17. Performance profiling to achieve peak network throughput, leveraging: one (1-ota) and two (2-ota) OTA UEs at a fixed location using iPerf and a DDDDDDSUUU TDD pattern; one (1-sim) and two (2-sim) emulated UEs using Keysight RuSIM and CoreSIM with a DDDDDDSUUU TDD pattern; and two (2-simdl, 2-simul) and twenty-five (25-simdl, 25-simul) emulated UEs with Keysight RuSIM and CoreSIM, a reduced number of guard symbols, a DDDDDDSUUU TDD pattern for DL cases, and a DDDSU TDD pattern for UL cases.

and the client algorithm Adaptive Bitrate Streaming (ABR) downgrades the bitrate based on the estimate it gets. This is due to a slow MCS selection loop in the OAI L2, which will be improved as part of our future work. However, this shows that our setup is capable of supporting up to 8K High Dynamic Range (HDR) videos that require 150 – 300 Mbps bitrates according to YouTube guidelines [42]. During mobility, the average bitrate is 120 Mbps, and the rebuffer ratio increases to 15%. This is once again because the UE moves away from the RU, gradually entering low-coverage regions and eventually disconnecting.

### E. Peak Performance Experiments

In this second set of experiments, we expand our evaluation to stress-test the system and attain peak performance results. To achieve these compared to previous tests, we leverage a GHRAN server with a DDDDDDSUUU TDD pattern, a 4x4 MIMO configuration, 4 layers DL, 1 layer UL and a  $Q_m$  up to 256-QAM. We compare system output with a single and double commercial OTA UEs connected to our network at a fixed location using Open5GS as CN and iPerf to generate traffic, as well as with the Keysight RuSIM emulator device, emulating both RU and up to 25 UEs, using Keysight CoreSIM to emulate the CN. The average throughput and 95% confidence intervals are plotted in Fig. 17. We observe that in OTA at a fixed location, the UEs achieve steady throughput in all cases, as indicated by the small confidence interval values, with a peak of up to 1.05 Gbps in DL and 100 Mbps in UL for a single UE (1-ota). Furthermore, the combined throughput increases with the number of connected UEs (2-ota), reaching a maximum of 1.2 Gbps in DL, while remaining close to 100 Mbps in UL.

By using the Keysight RuSIM emulator with the same configuration as OTA, performance improves to over 1.26 Gbps with a single UE (1-sim) and 1.42 Gbps with two UEs in DL (2-sim), and close to 110 Mbps in UL for both one and two UEs. This performance increase can be attributed to the more controlled environment provided by RuSIM, which eliminates external interference and impairments. In this case, an *ExcellentRadio-Conditions* channel model—also used for Base Station (BS) conformance testing as specified in the 3GPP specifications [43]—is enabled to simulate ideal radio conditions. To achieve the

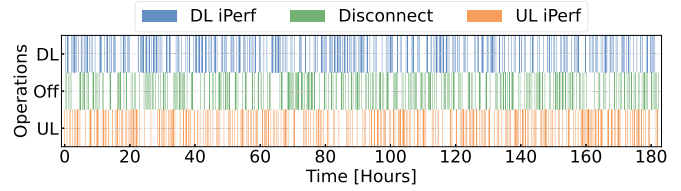


Fig. 18. Long-running stability experiment involving one UE randomly cycling for over 180 hours among three operations, repeated every 10 minutes: (blue) DL iPerf for 1 minute; (orange) UL iPerf for 1 minute; and (green) disconnection from the network for the remainder of the 10-minute cycle window.

current peak cell throughput, we leverage a DDDDDDSUUU TDD pattern in DL and a DDDSU pattern in UL, utilizing a reduced number of guard symbols (only one) enabled by RuSIM during two separate experiment runs with two emulated UEs. This approach results in an aggregate throughput of 1.68 Gbps in DL (2-simdl) and 143 Mbps in UL (2-simul). Moreover, we stress-test the system by simultaneously connecting up to 25 emulated UEs while exchanging traffic, achieving similar performance (25-simdl, 25-simul). This demonstrates that the network can reliably sustain multiple UEs and reaches its peak with two UEs, while fairly distributing resources when more devices are connected. These results highlight the maximum performance currently achievable by X5G, showcasing values comparable to those of production-level systems.

### F. Long-Running Experiments

To validate stability and reliability, we evaluate X5G through long-running experiments with a single UE performing continuous operations. The cell configuration remains the same as in Section V-E, utilizing the DDDDDDSUUU TDD pattern, a 4x4 MIMO setup with 4 DL layers and 1 UL layer. The UE is a Samsung S23 phone, which cycles randomly every 10 minutes between three different operations:

- DL test, a 1-minute UDP downlink iPerf data test targeting 50 Mbps.
- UL test, a 1-minute UDP uplink iPerf data test targeting 10 Mbps.
- Disconnection, the UE disconnects from the network, remains disconnected for the remaining 10 minutes, and then reconnects.

Fig. 18 shows the results of the long-running experiment, where the operations performed by the UE every 10 minutes are represented with colored bars. The system can sustain indefinite uptime, as highlighted in the figure with over 180 hours of operation before the cell was manually shut down to vacate the spectrum for other planned experiments in the area. Additionally, Fig. 19 presents some of the metrics available on the RAN server side, showing the resource utilization required to run the NVIDIA ARC gNB with a single cell on a GH200 GPU. Specifically, GPU utilization and power draw are depicted for a GH server during a one-hour window of the previous long-running stability experiment. We can see how the utilization (in blue) drops to nearly 10% when no UE is connected, and rises to approximately 50% during DL data traffic, reflecting the system's computation demand. On the other hand, during idle periods and UL communication, GPU utilization remains

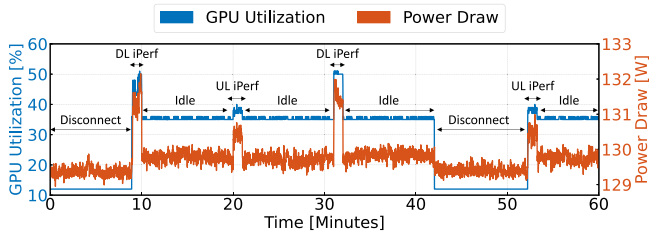


Fig. 19. GPU utilization (blue) and power draw (orange) of the NVIDIA Grace Hopper server node during a one-hour window of the long-running stability experiment. The results show the behavior of the system when the UE cycles through three operations: disconnecting for 10 minutes, performing a DL iPerf test for 1 minute, followed by 10 minutes of idling, and performing a UL iPerf test for 1 minute, followed by 10 minutes of idling.

stable between 35% and 40%, respectively. The power draw (in orange) follows a similar trend, ranging from 129 to 132 W. It is important to note that these results apply to a single cell, but the resource requirements for multiple cells do not scale linearly. Each GH server can support up to 20 cells [17] while maintaining a high-level of energy efficiency for RAN communications [44]. Overall, these results highlight the high reliability of X5G in terms of both performance and stability, positioning it as a suitable candidate for P5G deployments, as well as a valuable playground to develop, test, and evaluate novel AI/ML algorithms and solutions for the RAN.

## VI. RELATED WORK

This section compares the features and capabilities of the X5G testbed within the context of similar programmable open RAN and 5G, highlighting its unique features and contributions beyond 5G research and experimentation. Surveys of testbeds for open and programmable wireless networks can also be found in [3], [45].

The Platforms for Advanced Wireless Research (PAWR) [46] offers a set of geographically and technically diverse testbeds designed to enhance specific wireless communication areas. These include POWDER, AERPAW, COSMOS, ARA, and Colosseum, each equipped with specialized technologies to address varied research needs.

The POWDER facility, located at the University of Utah in Salt Lake City, UT, supports a wide spectrum of research areas, including next-generation wireless networks and dynamic spectrum access [47]. Its 5G stack is based primarily on a combination of open-source stacks, combined with Software-defined Radios (SDRs) or RUs but not accelerated at the physical layer, and on a commercial Mavenir system, which does not support access to the source code from the PHY to the core network, differently from the X5G stack.

Similarly, AERPAW, deployed on the campus of North Carolina State University in Raleigh, NC, focuses on aerial and drone communications, diverging from our emphasis on private 5G network configurations [48]. The AERPAW facility hosts an Ericsson 5G deployment with similar limitations with respect to stack programmability for research use cases.

The COSMOS project [49] leverages an array of programmable and software-defined radios, including USRP and

Xilinx RFSoc boards, to facilitate mmWave communication experiments across a city-scale environment. The outdoor facilities of COSMOS are deployed in the Harlem area, in New York City, while its indoor wireless facilities are on the Rutgers campus in North Brunswick, NJ. Unlike X5G, COSMOS is designed for broad academic and industry use and is more focused on mmWave deployments enabling diverse external contributions to its development without specific emphasis on any single network architecture.

The ARA testbed [50], deployed across Iowa State University (ISU), in the city of Ames, and surrounding rural areas in central Iowa, serves as a large-scale platform for advanced wireless research tailored to rural settings. ARA includes diverse wireless platforms ranging from low-UHF massive MIMO to mmWave access, long-distance backhaul, free-space optical, and Low Earth Orbit (LEO) satellite communications, utilizing both SDR and programmable COTS platforms and leveraging open-source software like OAI, srsRAN, and SD-RAN [51]. However, unlike the X5G testbed, ARA focuses primarily on rural connectivity without focusing on specialized hardware for PHY layer optimization or digital twin frameworks for RF planning.

Colosseum is the world's largest Open RAN digital twin [40], [52]. This testbed allows users to quickly instantiate software-defined cellular protocol stacks, e.g., the OAI one, on its 128 compute nodes. These nodes control 128 SDRs that are used as RF front-ends and are connected to a massive channel emulator, which enables experimentation in a variety of emulated RF environments. However, the Colosseum servers are not equipped to offload lower-layer cellular operations on GPUs, and the available SDRs are USRP X310 from NI, instead of commercial RUs.

The OSC is also involved in the creation of laboratory facilities [53] that comply with O-RAN standards and support the testing and integration of O-RAN-compliant components. These testing facilities, distributed across multiple laboratories, foster a diverse ecosystem through their commitment to open standards and collaborative development. However, unlike X5G, they do not explicitly focus on the deployment complexities of private networks, nor do they provide any PHY layer acceleration technology or utilize a digital twin for RF planning. Instead, they aim to promote multi-vendor interoperability within an open collaborative framework.

6G-SANDBOX [54] is a versatile facility that includes four geographically displaced platforms in Europe, each equipped to support a variety of advanced wireless technologies and experimental setups. It uses a mix of commercial solutions (for example, Nokia microcells, Ericsson Base Band Unit (BBU), and the Amarisoft stack) and open source solutions (for example, OAI and srsRAN) in diverse environments ranging from urban to rural settings. Unlike X5G, 6G-SANDBOX primarily facilitates wide-ranging 6G research through its extensive, multi-location infrastructure. Its predecessor, 5GENESIS [55], featured a modular and flexible experimentation methodology, supporting both per-component and end-to-end (E2E) validation of 5G technologies and KPI across five European locations. This testbed emphasizes a comprehensive approach to

5G performance assessment, integrating diverse technologies such as Software-defined Networking (SDN), Network Function Virtualization (NFV), and network slicing to enable rigorous testing of vertical applications but not including O-RAN architectures.

The Open AI Cellular (OAIC) testbed [56], developed at Virginia Tech, is an open-source 5G O-RAN-based platform designed to facilitate AI-based RAN management algorithms. It includes the OAIC-Control framework for designing AI-based RAN controllers and the OAIC-Testing framework for automated testing of these controllers. The OAIC testbed introduces a new real-time RIC, zApps, and a Z1 interface to support use cases requiring latency under 10 ms, integrated with the CORNET infrastructure for remote accessibility.

The CCI xG Testbed provides a comprehensive platform for advanced wireless research, particularly in the realm of 5G and beyond. It features a disaggregated architecture with multiple servers distributed across geographically disparate cloud sites, leveraging a combination of central and edge cloud infrastructures to optimize resource allocation and latency. The testbed includes several SDR-based CBRS Base Station Device (CBSD) integrated with an open-source Spectrum Access System (SAS) for dynamic spectrum sharing in the CBRS band [57]. Additionally, the testbed supports a full O-RAN stack using srsRAN and Open5GS and features both non-RT RIC and near-RT RIC for real-time and non-real-time radio resource management [58], [59].

The testbed in [60] provides a prototypical environment designed to experiment with vRAN deployments and evaluate resource allocation and orchestration algorithms. It focuses on the decoupling of radio software components from hardware to facilitate efficient and cost-effective RAN deployments. This testbed includes datasets that characterize computing usage, energy consumption, and application performance, which are made publicly available to foster further research. Unlike the X5G testbed, the O-RAN platform primarily addresses the flexibility and cost efficiency of virtualized RANs without incorporating specialized hardware for PHY layer tasks.

The disaggregated 5G testbed for live audio production [61] emphasizes ultra-reliable low-latency communication for media applications. Its scope is narrower than X5G, which supports a broader range of experimental scenarios and computationally intensive network configurations.

The data usage control framework [62] addresses privacy challenges in hybrid private-public 5G networks. While it highlights the importance of secure orchestration and policy management, its focus on analytics differs from X5G capabilities in physical layer acceleration and network performance experimentation.

Finally, the Microsoft enterprise-scale Open RAN testbed [63] highlights the potential of virtualized RAN functions on commodity servers, employing disaggregated architectures to demonstrate scalability and flexibility. By integrating Kubernetes for dynamic orchestration and using Intel FlexRAN with ACC100 accelerators for Low-Density Parity-Check (LDPC) look-aside offloading, this testbed achieves functional disaggregation of RAN workloads.

While state-of-the-art software stacks such as srsRAN already offer similar performance in terms of core 5G functionalities, including handovers, X5G distinguishes itself through its integration of GPU acceleration, enabling enhanced flexibility and computational power for future innovations. Unlike traditional platforms that rely on CPU-based architectures, which achieve performance parity for standard RAN tasks, it leverages GPUs not only for optimized PHY processing but also as a unified platform for AI/ML workloads. Indeed, the GPU architecture of X5G supports the development and deployment of dApps [13] that utilize AI/ML models for real-time network optimization. This capability aligns directly with the vision outlined by the AI-RAN Alliance [64], which emphasizes the integration of AI-driven decision-making processes across three key development areas: (i) AI-for-RAN, (ii) AI-and-RAN, and (iii) AI-on-RAN, making our platform an ideal candidate for advancing these areas. Moreover, the modular design of X5G guarantees compatibility with both open-source and commercial cores, facilitating future experiments with advanced technologies like massive MIMO, mmWave, and beamforming that are currently under development.

## VII. CONCLUSION AND FUTURE WORK

We introduced X5G, an open, programmable, and multi-vendor private 5G O-RAN testbed deployed at Northeastern University in Boston, MA. We demonstrated the integration of NVIDIA Aerial, a PHY layer implementation on GPUs, with higher layers based on OAI, resulting in the creation of the NVIDIA ARC platform. We provided an overview of the ARC software and hardware implementations, designed for a multiple-node deployment, including a Red Hat OpenShift cluster for the OSC RIC deployment, as well as examples of a KPM xApp and a slicing xApp. Additionally, we conducted a ray-tracing study using our digital twin framework to determine the optimal placement of X5G RUs. Finally, we discussed platform performance with varying numbers of COTS and emulated UEs and applications, such as iPerf and video streaming, as well as through long-running and stress-test experiments to evaluate its stability.

Next, we plan to continue the deployment of X5G gNBs comprising a mix of indoor and outdoor locations for more realistic experiments and comprehensive development of UE handover procedures. We are targeting the integration of RUs from different vendors and supporting bands for 5G New Radio (NR) Frequency Range 2 (FR2). We will also develop pipelines for the automatic deployment, testing, and management of workloads leveraging the Red Hat OpenShift cluster already in use for the OSC RIC integration. Our aims include (i) deploying a fully functional and reliable private 5G network that remains continuously up and running, providing an infrastructure for users to operate on and for researchers to collect realistic datasets, and (ii) enabling full RAN control to facilitate dynamic changes in network behavior by enhancing the capabilities of X5G, thereby offering the research community an end-to-end open and programmable platform for the development and testing of next-generation wireless networks and algorithms.

## ACKNOWLEDGMENT

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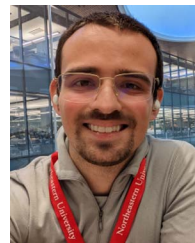
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5G standalone access, in 2021. He is currently on sabbatical with Northeastern University, where he is working on bringing different open-source communities around 5G and open RAN together to build an end-to-end reference architecture for 6G research.

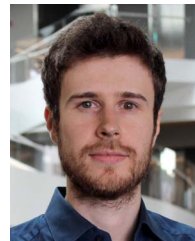
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Qualcomm in the connected cars (CV2X) ecosystem. And if you ever fly JetBlue, United, Continental, or Quantas, she was instrumental in the inflight broadband connectivity over satellite networks.



**Chris Dick** joined NVIDIA, in 2020 where he is a system architect working on the application of Artificial Intelligence and Machine Learning to 5G and 6G wireless. In his 30 years working in signal processing and communications he has delivered silicon and software products for 3G, 4G, and 5G baseband DSP and Docsis 3.1 cable access and vector processor architectures. He has performed research and delivered products for digital front-end (DFE) technology for cellular systems with a particular emphasis on digital pre-distortion for power amplifier linearization. He has also worked extensively on silicon architecture and compilers for machine learning and parallel computing architectures. Prior to moving to Silicon Valley, in 1998, he was a tenured academic in Melbourne Australia for 13 years. He has more than 250 publications and 100 patents. From 1998 to 2020 he was a fellow and the DSP chief architect with Xilinx. In 2018 he was awarded the IEEE Communications Society Award for Advances in Communication for research in the area of full-duplex wireless communication.



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of Nano-Things. In these areas, he has co-authored more than 250 peer-reviewed scientific publications, including one book, and has been granted five US patents. His work has received more than 17,000 citations (h-index of 61 as of June 2024). He is serving as the lead PI on multiple grants from U.S. federal agencies including the National Science Foundation, the Air Force Office of Scientific Research, and the Air Force Research Laboratory as well as industry. He is the recipient of multiple awards, including the NSF CAREER Award, in 2019, the 2022 IEEE ComSoc RCC Early Achievement Award, and the 2022 IEEE Wireless Communications Technical Committee Outstanding Young Researcher Award, among others, as well as four best paper awards. He is a *IEEE ComSoc distinguished lecturer* (2022–2024). He is also the editor-in-chief of the Elsevier *Nano Communication Networks Journal* and editor for *IEEE Transactions on Communications* and *Nature Scientific Reports*.



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